Capstone Project 3: Dashboard Development

Assignment Type: Individual

Submission Format: Report + Dashboard Files + SQL Queries

Due Date: June 7th 23:59

Objective:

The goal of this project is to develop two interactive dashboard using the cleaned dataset and database from Capstone Project 2. This dashboard should provide meaningful insights, visualizations, and an intuitive user experience. You will learn how to integrate front-end, back-end, and database elements to create a data-driven application.

Project Guidelines:

Step 1: Define Dashboard Requirements

- **Understand the end users** – Who will use the dashboard? What are their demographics?
- **Identify key metrics** – What data insights are most important?
- **Choose appropriate charts** – What types of visualizations best represent the data?
- **Story telling** – What is the story behind your dashboard?
- **Dashboard type** – What type of dashboard are you going to design to tell the story? Analytical, Operational, Strategic, Managerial (Tactical). Both dashboards need to be different types.

Deliverable:

- A brief **Dashboard Plan** outlining story, type of dashboard, key features, target users, and required visualizations.

Step 2: Database Integration

- Connect the **database from Capstone Project 2** to the dashboards.
- Optimize queries for **data retrieval**.

- Perform aggregations and filtering for dashboard performance.

Deliverable:

- **SQL Queries used for data retrieval and transformation.**
- **Database connection code (Python/Flask, SQLAlchemy, etc.).**

Step 3: Dashboard Design

- **Define the layout and UX:**
  - Widgets
  - Logical placement of charts, filters, and tables.
  - Interactive elements (dropdowns, search, date filters).
- **Tools:**
  - Python (Flask compulsory)
  - JavaScript (Chart.js, D3.js)
  - ObservableHQ (optional addition)
  - Tableau / Power BI (optional addition)

Deliverable:

- **N/A**

Step 4: Develop the Dashboard

- **Front-End:**
  - Use **HTML, CSS, JavaScript** for styling and responsiveness.
  - Implement interactive elements using **Flask, and JavaScript libraries**.
- **Back-End:**
  - Connect the database using Flask.
  - Fetch data dynamically and render it on the dashboard.

Deliverable:

- **2 Fully functional dashboard.**
- **Screenshots of the working dashboard.**
- **Source code files (HTML, CSS, JavaScript, Python, etc.) on GitHub link**

Step 5: Data Visualization

- Implement **charts, graphs, tables** to present data insights.
- Ensure **real-time data updates** if applicable.
- Optimize the dashboard for **performance and readability**.

Deliverable:

- **At least 4-5 meaningful visualizations per dashboard.**

- **At least 4 different quantitative visualization techniques need to be used**
- **At least 3 Qualitative Visualization Techniques Need to be Used**
- **Interactive elements such as filters and search.**

Step 6: User Experience (UX) and Testing

- **Test dashboard usability:**
  - Is it intuitive and user-friendly?
  - Are the charts easy to interpret?
  - Does it load efficiently?
- **Gather feedback** from users or peers.
- **Fix bugs** and improve dashboard responsiveness.

Deliverable:

- **List of UX considerations and improvements fully integrated into the dashboard screenshots from step 5**
- **Screenshots of dashboard functionality.**

Submission Requirements

Your final submission should include:

1. **Dashboard Plan (Step 1)**
2. **Database Connection & SQL Queries (Step 2)**
3. **N/A (Step 3)**
4. **Dashboard Code & Screenshots (Step 4-5)**
5. **Screenshots of Dashboard (Step 6)**

Evaluation Criteria (Rubric)

Category	Points	Description
Dashboard Planning	15	Clear identification of goals, audience, and key metrics for both dashboards. No story, no marks. Different dashboard types required.
Database Integration	20	Efficient database connection and optimized queries for two dashboards.
Dashboard Design & UX	20	Well-structured layout and intuitive user experience for both dashboards.
Functional Dashboard	25	Working dashboard with interactive elements and visualizations for both dashboards.
Data Visualization	20	Meaningful charts, graphs, and tables to represent insights for both dashboards. At least 4 quantitative. At least 3 qualitative

Total: 100 points